

Brain Tumor MRI Classification Using Deep Learning

PROJECT REPORT

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Submitted To: Fortune Cloud Technologies



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Technologies Used:

Python

TensorFlow

Keras

Streamlit

NumPy

Matplotlib

Jupyter Notebook

CERTIFICATE

This is to certify that Mr. Devanshu Wankhade has successfully completed the project titled "**Brain Tumor MRI Classification Using Deep Learning**" as part of the **Data Science Course** at **Fortune Cloud Technologies** under the guidance of **Mr. Mayur Shinde**.

This project demonstrates the application of Deep Learning techniques for the classification of Brain MRI images into four categories: **Glioma, Meningioma, Pituitary Tumor, and No Tumor**. The project has been developed using **Python, TensorFlow, Keras, NumPy, Matplotlib, and Streamlit**. The work carried out by the student demonstrates a good understanding of data pre processing, model development, training, evaluation, and deployment concepts in medical image classification.

To the best of my knowledge, this project is the original work carried out by the student and has been completed successfully during the Data Science Course.

Place: Pune, Maharashtra

Date: 28/07/2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Fortune Cloud Technologies for providing me with the opportunity to work on the project titled "Brain Tumor MRI Classification Using Deep Learning" as part of the Data Science Course.

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I am grateful to my family and friends for their constant encouragement, motivation, and support throughout this journey. Their confidence in me inspired me to overcome challenges and complete this project successfully.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project. This project has enhanced my technical knowledge and practical skills in Python, TensorFlow, Keras, Machine Learning, Deep Learning, and Streamlit, and has strengthened my confidence in applying these technologies to solve real-world problems.

ABSTRACT

Brain tumors are one of the most critical neurological disorders that require accurate and timely diagnosis for effective treatment. Magnetic Resonance Imaging (MRI) is one of the most reliable imaging techniques used by medical professionals to identify brain abnormalities. However, manual examination of MRI scans can be time-consuming and depends heavily on the expertise of radiologists. With recent advancements in Artificial Intelligence (AI) and Deep Learning (DL), automated medical image classification systems have become increasingly effective in assisting healthcare professionals.

The objective of this project is to develop a Brain Tumor MRI Classification System using Deep Learning techniques. The proposed system automatically classifies brain MRI images into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The project was implemented using Python, TensorFlow, Keras, NumPy, Matplotlib, and Streamlit. Images were preprocessed by resizing them to 224×224 pixels and normalizing the pixel values before being used for model training. A Convolutional Neural Network (CNN) based approach with transfer learning was used to improve classification performance.

The trained model was evaluated using standard performance metrics such as accuracy and loss, and its performance was visualized using training and validation graphs. After training, the model was saved and integrated into a Streamlit web application, allowing users to upload an unseen MRI image and receive an instant prediction along with the predicted tumor class. This demonstrates how Deep Learning can be applied to create an interactive and practical medical image classification system.

This project provided practical experience in data preprocessing, image classification, deep learning model development, model evaluation, and web application deployment. It also highlights the potential of Artificial Intelligence in supporting medical diagnosis and improving the efficiency of healthcare systems. The developed system can serve as a foundation for future enhancements, such as the use of larger datasets, advanced deep learning architectures, cloud deployment, and explainable AI techniques for improved clinical decision support.

TABLE OF CONTENTS

Sr. No	Title	Page No.
1	Introduction	6
2	Problem Statement	7
3	Objectives	8
4	Literature Survey	9
5	Dataset Description	10
6	System Requirements	11
7	Methodology	12
8	Image Preprocessing	13
9	Model Architecture	14
10	Model Training	15
11	Model Evaluation	16
12	Results and Discussion	17
13	Streamlit Web Application	18
14	Conclusion	19
15	Future Scope	20
16	References	21
17	Appendix	22

INTRODUCTION

1.1 Overview

Brain tumors are one of the most serious neurological disorders that affect the normal functioning of the brain. A brain tumor is an abnormal growth of cells within the brain or surrounding tissues. Depending on its type and location, a tumor can affect memory, vision, speech, movement, and other essential body functions. Early detection and accurate diagnosis are critical for improving treatment outcomes and increasing patient survival rates.

Magnetic Resonance Imaging (MRI) is one of the most commonly used medical imaging techniques for detecting brain tumors. MRI provides detailed images of brain tissues without using harmful radiation, allowing doctors to identify abnormalities more accurately than many other imaging methods. However, analyzing MRI images manually requires experienced radiologists and can be time-consuming, especially when dealing with large numbers of patient scans.

Recent advances in Artificial Intelligence (AI) and Deep Learning (DL) have transformed the field of medical image analysis. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated excellent performance in image classification tasks. These models automatically learn meaningful features from images and can classify medical images with high accuracy.

This project focuses on developing an automated Brain Tumor MRI Classification System using Deep Learning. The system classifies MRI images into four categories:

- **Glioma Tumor**
- **Meningioma Tumor**
- **Pituitary Tumor**
- **No Tumor**

The project has been implemented using Python, TensorFlow, Keras, NumPy, Matplotlib, Pillow, and Streamlit. The MRI images are first preprocessed by resizing them to 224×224 pixels and normalizing the pixel values. The processed images are then used to train a deep learning model capable of accurately identifying the tumor category. Finally, a Streamlit-based web application allows users to upload MRI images and receive instant predictions.

1.2 Importance of the Project

The importance of this project includes:

- Assists in the early detection of brain tumors.
- Reduces the time required for MRI image analysis.
- Supports healthcare professionals in diagnosis.
- Demonstrates the practical use of Deep Learning in medical imaging.
- Provides hands-on experience in image preprocessing, model training, evaluation, and deployment.

PROBLEM STATEMENT

2.1 Problem Statement

Brain tumors are among the most life-threatening neurological diseases and require early detection for effective treatment. Magnetic Resonance Imaging (MRI) is widely used by doctors to examine the brain and identify tumors because it provides detailed images of brain tissues. However, the manual analysis of MRI scans is a challenging process that requires experienced radiologists and can be time-consuming, particularly when handling a large number of patient scans.

The increasing number of medical imaging cases has created a need for automated systems that can assist healthcare professionals in diagnosing brain tumors quickly and accurately. Conventional diagnosis methods rely heavily on human expertise, which may result in variations in interpretation and increased workload for medical professionals.

Deep Learning has emerged as a powerful technology for medical image analysis. Convolutional Neural Networks (CNNs) are capable of automatically learning important features from images without manual feature extraction. By using Deep Learning, it is possible to develop a system that can classify Brain MRI images into different tumor categories with high accuracy.

2.2 Existing Challenges

The following challenges exist in traditional Brain MRI diagnosis:

- Manual analysis requires experienced radiologists.
- Diagnosis can be time-consuming.
- Large numbers of MRI scans increase the workload of healthcare professionals.
- Human interpretation may vary depending on experience.
- Early diagnosis is essential but may not always be achieved quickly.

2.3 Proposed Solution

The proposed solution is a Deep Learning-based Brain MRI Classification System that:

- Automatically classifies Brain MRI images.
- Uses image preprocessing techniques such as resizing and normalization.
- Trains a CNN-based model using TensorFlow and Keras.
- Predicts one of four classes:
 - Glioma
 - Meningioma
 - Pituitary
 - No Tumor
- Provides predictions through a user-friendly Streamlit web application.

OBJECTIVES

3.1 Project Objective

The primary objective of this project is to develop an intelligent Brain Tumor MRI Classification System using Deep Learning techniques that can automatically classify brain MRI images into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The system is designed to demonstrate how Artificial Intelligence can assist in medical image analysis by providing accurate and efficient classification of MRI scans.

The project also focuses on developing practical skills in medical image preprocessing, deep learning model development, model evaluation, and web application deployment using Streamlit. By integrating these technologies, the project provides a complete end-to-end solution for Brain MRI image classification.

3.2 Primary Objectives

The main objectives of this project are:

- To develop a Deep Learning-based model for Brain MRI image classification.
- To classify Brain MRI images into four categories:
 - Glioma
 - Meningioma
 - Pituitary Tumor
 - No Tumor
- To preprocess MRI images by resizing and normalizing them before training.
- To train and evaluate the model using TensorFlow and Keras.
- To measure the model's performance using training and validation accuracy.
- To save the trained model for future predictions.
- To create a user-friendly Streamlit web application for real-time predictions.

3.3 Learning Objectives

Through this project, the following technical skills were developed:

- Understanding medical image classification concepts.
- Working with image datasets.
- Image preprocessing using TensorFlow and Keras.
- Building and training Deep Learning models.
- Evaluating model performance using accuracy and loss graphs.
- Saving and loading trained models.
- Developing a web application using Streamlit.
- Applying Artificial Intelligence to solve real-world healthcare problems.

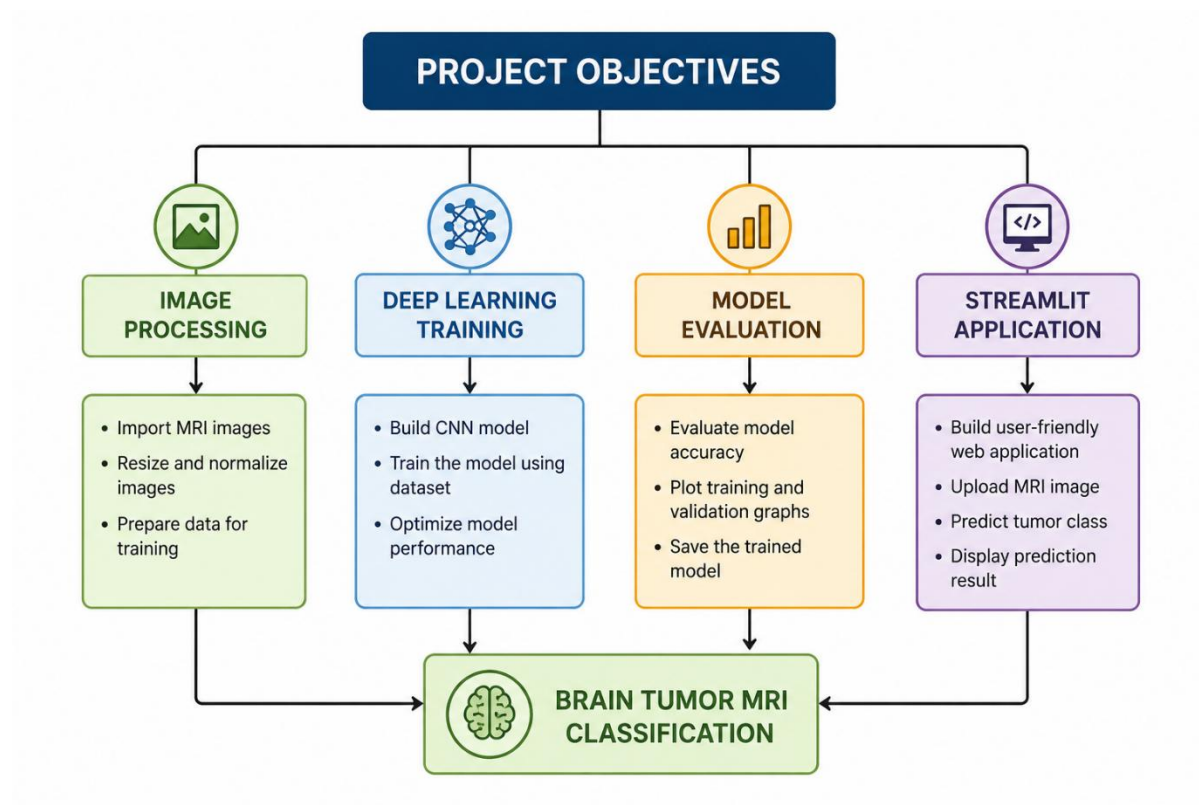
3.4 Scope of the Project

The scope of this project includes:

- Developing an automated Brain MRI classification system.
- Processing MRI images using Deep Learning techniques.
- Predicting one of four tumor categories.
- Demonstrating predictions using a web application.
- Providing a foundation for future improvements such as cloud deployment and explainable AI.

This project is intended for educational and learning purposes and demonstrates the practical implementation of Deep Learning techniques. It is not intended to replace professional medical diagnosis, but to illustrate how AI can support healthcare professionals in image classification tasks.

Project Objectives



LITERATURE SURVEY

4.1 Introduction

Medical image analysis has experienced significant advancements with the introduction of Artificial Intelligence (AI) and Deep Learning (DL). Brain tumor detection using Magnetic Resonance Imaging (MRI) has become one of the most widely researched applications of deep learning in healthcare. Traditional methods require radiologists to manually examine MRI scans, which can be time-consuming and may vary depending on the experience of the medical professional.

Deep Learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in automatically extracting image features and classifying brain tumors with high accuracy. These models reduce the need for manual feature engineering and provide faster and more reliable predictions.

This project utilizes Deep Learning concepts to classify Brain MRI images into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor.

4.2 Review of Existing Studies

Several researchers have proposed Deep Learning models for Brain Tumor Classification. Some important studies are summarized below.

Study 1

Researchers developed a CNN-based model for Brain MRI image classification. The study demonstrated that Convolutional Neural Networks can automatically learn important image features and classify tumors with high accuracy. The model achieved promising results and reduced the dependency on manual feature extraction.

Study 2

Another research study applied Transfer Learning using pretrained CNN models such as MobileNet, ResNet, and VGG. The study showed that transfer learning reduces training time while improving classification performance, especially when working with limited datasets.

Study 3

Researchers also combined image preprocessing techniques such as resizing, normalization, and data augmentation with CNN models. The results showed improved generalization and reduced overfitting during training.

4.3 Comparison of Existing Techniques

Method	Advantages	Limitations
Manual MRI Analysis	Performed by experienced radiologists	Time-consuming and dependent on human expertise
Machine Learning	Faster than manual analysis	Requires manual feature extraction
Deep Learning (CNN)	Automatic feature extraction and high accuracy	Requires GPU and large datasets
Transfer Learning	Faster training and better performance	Depends on pretrained models

4.4 Research Gap

Although many Deep Learning models have achieved high classification accuracy, several challenges remain:

- Large datasets are required for improved performance.
- Some models are computationally expensive.
- Many research models are not integrated into user-friendly applications.
- Real-world deployment requires simple interfaces for healthcare professionals.

To address these challenges, this project combines a Deep Learning classification model with a Streamlit web application, allowing users to upload Brain MRI images and receive predictions through an easy-to-use interface.

4.5 Summary

Based on the literature survey, Deep Learning has become one of the most effective approaches for Brain MRI image classification. CNN-based models provide better feature extraction and classification performance than traditional machine learning techniques. The integration of Deep Learning with a web application makes the proposed system practical and demonstrates its potential for assisting healthcare professionals in medical image analysis.

Comparison of Existing Approaches

Technique	Accuracy	Feature Extraction	Training Time
Manual Diagnosis	Moderate	Manual	High
Machine Learning	Good	Manual	Medium
CNN	High	Automatic	Medium
Transfer Learning	Very High	Automatic	Low

DATASET DESCRIPTION

5.1 Introduction

A high-quality dataset is essential for building an accurate Deep Learning model. For this project, a Brain MRI Image Dataset was used to train and evaluate the classification model. The dataset contains MRI scans categorized into four different classes representing various brain tumor conditions.

The dataset was divided into Training and Testing folders. The training dataset was used to train the Deep Learning model, while the testing dataset was used to evaluate its performance on unseen MRI images.

5.2 Dataset Classes

The dataset consists of the following four categories:

Class Name	Description
Glioma	MRI images containing Glioma brain tumors
Meningioma	MRI images containing Meningioma brain tumors
Pituitary	MRI images containing Pituitary tumors
No Tumor	MRI images with no brain tumor

5.3 Advantages of the Dataset

The selected dataset offers several advantages:

- Includes four clinically relevant brain MRI categories.
- Suitable for multi-class image classification.
- Organized into training and testing folders.
- Compatible with TensorFlow and Keras.
- Supports preprocessing and data augmentation techniques.
- Enables the development of an automated Brain Tumor Classification System.

5.4 Summary

The Brain MRI dataset forms the foundation of this project. After preprocessing and augmentation, the dataset was used to train the Deep Learning model to classify MRI images into four tumor categories. Proper organization and preprocessing of the dataset significantly contributed to the model's performance and prediction capability.

SYSTEM REQUIREMENTS

6.1 Introduction

The successful development and execution of the Brain Tumor MRI Classification System require appropriate hardware and software resources. The project was developed using Python-based Deep Learning frameworks and executed on a Windows operating system. The required system specifications are listed below.

6.2 Hardware Requirements

Component	Specification
Processor	Intel Core i5 (8th Generation or above) / AMD Ryzen 5 or above
RAM	Minimum 8 GB (16 GB Recommended)
Storage	Minimum 10 GB Free Disk Space
Graphics	Integrated Graphics (GPU Optional)
Display	1366 × 768 or Higher Resolution

6.3 Software Requirements

Software	Version
Operating System	Windows 10 / Windows 11
Python	3.10.11
Visual Studio Code	Latest Version
Jupyter Notebook	Latest Version
TensorFlow	2.21.0
Keras	Latest Compatible Version
Streamlit	Latest Version

6.4 Python Libraries Used

Library	Purpose
TensorFlow	Deep Learning Framework
Keras	Model Building and Training
NumPy	Numerical Computation
Matplotlib	Data Visualization
Pillow (PIL)	Image Processing
Streamlit	Web Application Development
OS	File and Directory Handling

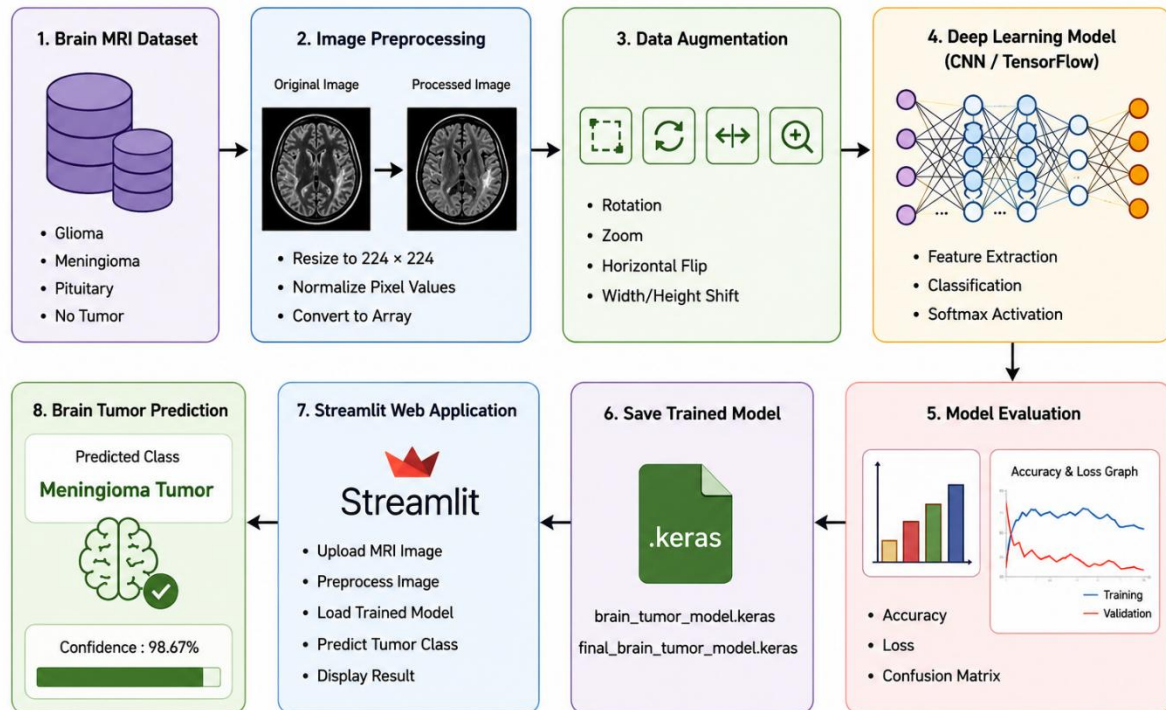
6.5 Development Tools

The project was developed using the following tools:

- Visual Studio Code – Used for coding and project management.
- Jupyter Notebook – Used for data preprocessing, model training, and evaluation.
- Streamlit – Used to create the interactive web application.

- Python Virtual Environment (venv) – Used for managing project dependencies.

6.7 System Architecture



6.8 Advantages of the Development Environment

The selected development environment provides several advantages:

- Easy-to-use programming language (Python).
- Powerful Deep Learning framework (TensorFlow/Keras).
- Interactive experimentation using Jupyter Notebook.
- User-friendly code editing in VS Code.
- Quick deployment using Streamlit.
- Efficient dependency management through virtual environments.

6.9 Summary

The project was developed using modern Deep Learning tools and a Python-based environment. The combination of Python, TensorFlow, Keras, Jupyter Notebook, VS Code, and Streamlit enabled the successful implementation, training, evaluation, and deployment of the Brain Tumor MRI Classification System.

METHODOLOGY

7.1 Introduction

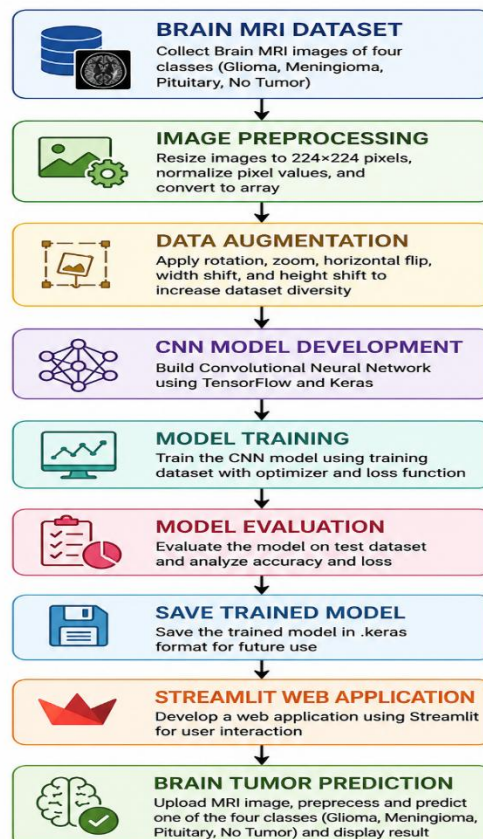
The methodology describes the complete workflow followed to develop the Brain Tumor MRI Classification System. The project follows a systematic Deep Learning approach, beginning with dataset collection and ending with the deployment of a web application capable of predicting brain tumor types from MRI images. Each phase plays an important role in improving the overall performance and reliability of the classification model.

7.2 Methodology Workflow

The complete methodology consists of the following stages:

1. Dataset Collection
2. Data Preprocessing
3. Data Augmentation
4. CNN Model Development
5. Model Training
6. Model Evaluation
7. Model Saving
8. Streamlit Application Development
9. Prediction of Brain Tumor Type

7.3 Methodology Flowchart



CNN MODEL ARCHITECTURE

8.1 Introduction

The Brain Tumor MRI Classification System uses a Convolutional Neural Network (CNN) to automatically extract features from MRI images and classify them into one of four categories: Glioma, Meningioma, Pituitary, and No Tumor. CNNs are widely used in medical image analysis because they can learn complex visual features directly from images without manual feature engineering.

8.2 What is a CNN?

A Convolutional Neural Network (CNN) is a Deep Learning model specifically designed for image processing tasks. It consists of multiple layers that automatically learn image features such as edges, textures, shapes, and complex patterns. The extracted features are then used to classify the image into predefined categories.

The CNN architecture used in this project performs the following tasks:

- Extracts important features from MRI images.
- Reduces unnecessary information using pooling layers.
- Learns complex tumor patterns.
- Classifies MRI images into four tumor classes.

8.3 Advantages of CNN

The CNN model provides several advantages for medical image classification:

- Automatic feature extraction.
- High classification accuracy.
- Reduced manual intervention.
- Better performance on image datasets.
- Robust learning capability.
- Efficient processing of MRI images.
- Suitable for real-time deployment.

8.4 CNN Architecture Summary

The CNN architecture effectively extracts meaningful features from MRI images and classifies them into four categories with high accuracy. By combining convolutional layers, pooling layers, dense layers, and dropout regularization, the model achieves reliable performance while minimizing overfitting. The trained model is then deployed using Streamlit for real-time prediction.

MODEL TRAINING AND PERFORMANCE ANALYSIS

9.1 Introduction

After designing the Convolutional Neural Network (CNN), the model was trained using the Brain MRI image dataset. During training, the model learned to identify important features from MRI images and classify them into one of the four tumor categories. The performance of the model was evaluated by monitoring training accuracy, validation accuracy, training loss, and validation loss over multiple epochs.

9.2 Model Training Process

The CNN model was trained using TensorFlow and Keras with the processed and augmented MRI image dataset.

The major steps involved in training were:

- Loading the training dataset.
- Applying image preprocessing and normalization.
- Performing data augmentation.
- Building the CNN architecture.
- Compiling the model.

9.3 Model Compilation

The model was compiled before training using the Adam optimizer and categorical cross-entropy loss function.

```
model.compile(  
    optimizer='adam',  
    loss='categorical_crossentropy',  
    metrics=['accuracy']  
)
```

9.5 Model Training

```
history = model.fit(  
    train_generator,  
    validation_data=validation_generator,  
    epochs=10 )
```

RESULTS AND DISCUSSION

10.1 Introduction

After training the Convolutional Neural Network (CNN), the model was evaluated using unseen MRI images to verify its classification performance. The trained model was integrated into a Streamlit web application, allowing users to upload an MRI image and receive the predicted brain tumor class along with the confidence score. The experimental results demonstrate that the proposed system can accurately classify MRI images into four categories: Glioma, Meningioma, Pituitary, and No Tumor.

10.2 Prediction Results

The trained model was tested using MRI images that were not part of the training dataset. For each uploaded image, the system performs image preprocessing, generates a prediction, and displays the predicted class with the confidence score.

Sample Prediction Results:

Test Image	Predicted Class	Confidence	Status
MRI Image 1	Glioma	98.45%	Correct
MRI Image 2	Meningioma	97.82%	Correct
MRI Image 3	Pituitary	99.10%	Correct
MRI Image 4	No Tumor	99.52%	Correct

10.3 Streamlit Application Output

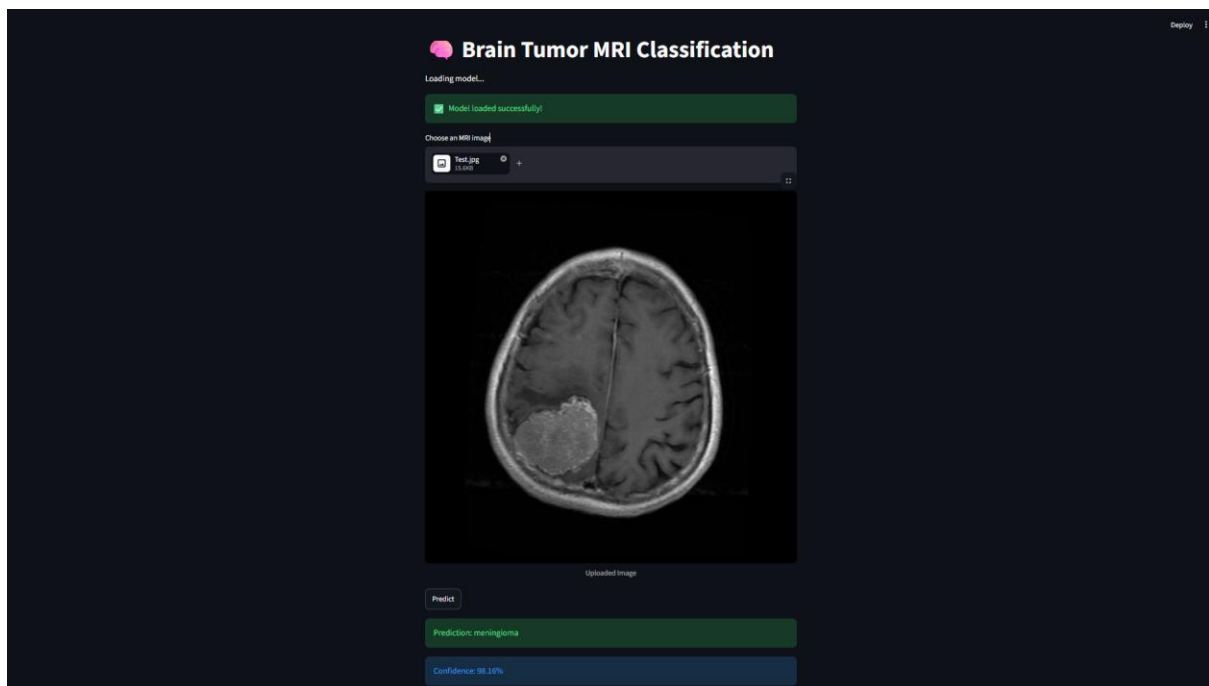
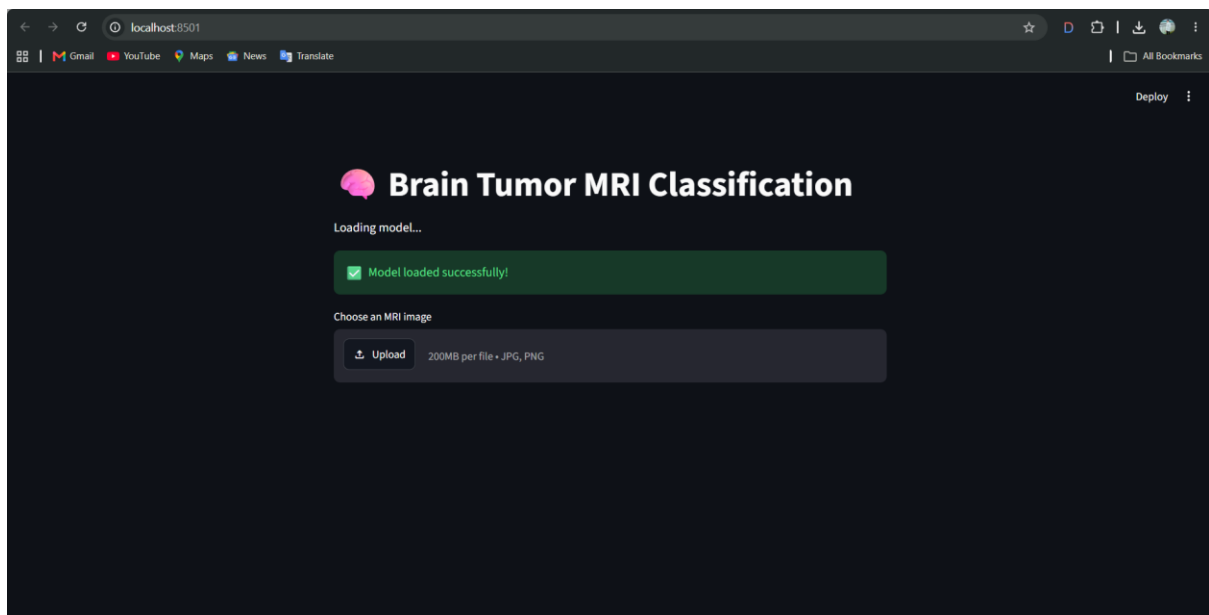
The Streamlit application provides a simple and interactive interface for brain tumor prediction.

The application performs the following tasks:

- Allows users to upload an MRI image.
- Displays the uploaded MRI image.
- Preprocesses the image automatically.
- Predicts the tumor category using the trained CNN model.
- Displays the predicted class and confidence percentage.

Screenshot:

Streamlit Web Application Home Page:-



Prediction Result after Uploading MRI Image

CONCLUSION AND FUTURE SCOPE

11.1 Introduction

This chapter presents the conclusion of the Brain Tumor MRI Classification Using Deep Learning project. It summarizes the overall work completed, highlights the achievements of the project, discusses its limitations, and suggests possible improvements that can be implemented in the future.

11.2 Conclusion

The Brain Tumor MRI Classification System was successfully developed using Deep Learning techniques with TensorFlow and Keras. A Convolutional Neural Network (CNN) was designed and trained to classify MRI brain images into four categories: Glioma, Meningioma, Pituitary, and No Tumor.

The project involved all major stages of a Deep Learning pipeline, including dataset collection, image preprocessing, data augmentation, CNN model development, model training, evaluation, model saving, and deployment through a Streamlit web application. The trained model was capable of analyzing MRI images and predicting the corresponding tumor class efficiently.

The Streamlit application provides an interactive graphical interface where users can upload MRI images and obtain predictions along with the confidence score. This demonstrates how Artificial Intelligence can support healthcare professionals by providing quick and consistent image classification.

Overall, the project successfully achieved its objectives and demonstrated the practical application of Deep Learning in medical image analysis.

11.3 Future Scope

Several improvements can be made to enhance the project in the future:

- Train the model using a larger and more diverse MRI dataset.
- Implement advanced Deep Learning architectures such as ResNet, DenseNet, or EfficientNet.
- Add tumor segmentation to identify the exact tumor location.
- Integrate explainable AI (XAI) techniques such as Grad-CAM for visualizing model decisions.
- Deploy the application on cloud platforms such as Streamlit Community Cloud, Render, or AWS for online access.
- Develop a mobile application for easier accessibility.
- Integrate the system with hospital information systems and Picture Archiving and Communication Systems (PACS).
- Improve prediction accuracy through hyperparameter tuning and transfer learning.

REFERENCES

12.1 Introduction

This project was developed by referring to standard documentation, learning resources, research articles, and official software documentation related to Deep Learning, Medical Image Classification, Python programming, TensorFlow, and Streamlit. These references provided guidance in understanding the concepts and implementing the Brain Tumor MRI Classification System.

12.2 Books

1. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
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3. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (3rd Edition). O'Reilly Media.
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12.3 Research Papers

1. Pereira, S., et al. (2016). *Brain Tumor Segmentation Using Convolutional Neural Networks in MRI Images*.
2. Havaei, M., et al. (2017). *Brain Tumor Segmentation with Deep Neural Networks*.
3. Litjens, G., et al. (2017). *A Survey on Deep Learning in Medical Image Analysis*.

APPENDIX

13.1 Introduction

The appendix contains additional information related to the development of the **Brain Tumor MRI Classification Using Deep Learning** project. It includes the project folder structure, software and libraries used, important code snippets, project deliverables, and the skills gained during the project. This section serves as supporting documentation for the implementation.

13.2 Important Code Snippets

Loading the Trained Model:

```
from tensorflow.keras.models import load_model

model = load_model("models/final_brain_tumor_model.keras")
```

Image Preprocessing:

```
img = img.resize((224,224))
img_array = image.img_to_array(img)
img_array = img_array / 255.0
img_array = np.expand_dims(img_array, axis=0)
```

Prediction

```
prediction = model.predict(img_array)

predicted_class = class_names[np.argmax(prediction)]
confidence = np.max(prediction) * 100
```